Theory of Computation

Revlew of Mathematical Theory

1-7 Let us discuss each one by one -

l) Negation

This proposition is also called NOT proposition. It is denoted by I or ~ or having horizontal bar on letter.

If S is a proposition then S defines NOr s. The truth table for this connective is

	1s
T	F-
F	1

Conjunction (AND)

If A and B are two propositions then conjunction of A and B is denoted by A^B. The truth table for this connective is

A	B	A^B
T	T	T
T	F	F
F	T	F
F	.F	F

lii) Disjunction OR)

If A and B are two propositions then disjunction of A and B is denoted by AvB. The truth table for this connective is

A	B	AvB
T	T	T
T	F	T
F	T	T
F	F	F

iv) Implication

If A and B are two propositions then if A then B is a proposition. It can be denoted byA-BorA = B.

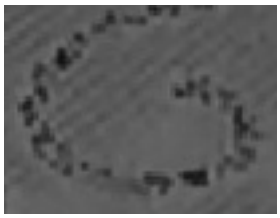


Figure 1

Roviow of Mathematical Theory

1-8

Theory of Computation

The truth table for this connective is

AB B A

T T T

F F T

T T F

T F F

For example Find truth value for following proposition.

If an apple is not green then it must be red".

Let. A denotes "An apple is not green" and B denotes It must be red The statement A is false and B is true. Hence the proposition is true.

v) If and only if

If A and B are two propositions then A if and only f B. It can be denoted as AB

The truth table will be

A	B	AB
T	T	T
T	F	F
F	T	F
F	F	T

For example: I will go to the school if and only if it is not raining Here statement A is I will go to the school". The statement B is It is not raining. Thus the resultant proposition is false.

1.5 Relations

Relaonship is a major aspect between two objects, even this is true in our real life One ohfect can be related with the other object by a some relation. Then those two objects forma pair based on this certain relationship.

Definition The relation R is a colection for the set S which represents the pair of elements,

For example (a b is in R We can represent their relation as a R b. The first component of each pair is chosen from a set called domain and second component of each pair is chosen from a set called range.

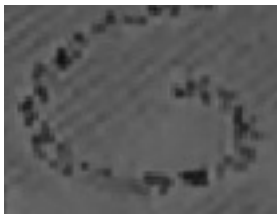


Figure 1

Review of Mathomatical Theory

Theory of Computation

1-9

Properties of Relations

A relation R on set S is

1. Reflexive if iRi for all i in S.
2. Irreflexive if iRi is false for all i in S.
3. Transitive if iRj and jRk imply iRk
4. Symmetric if iRj implies jRi.
5. Asymmetric if iRj implies that jRi is false.

Every asymmetric relation must be irreflexive.

For example: ifA = a. b then

reflexive relation R can be = a, ab, b

irreflexive relation R can be =a, b)

transitive relation R can be = a. b). b, aa. a

symmetric relation R can be=|a b. ba

asymmetric relation R can be =a, b

Equivalent Relation

A relation is said to be equivalence relation if it is reflexive, symmetric and transitive, over some set S.

Suppose R is a set of relations and S is the set of elements. For example S is the set of lines in a plane and R is the relation of lines intersecting to each other.

A relation is said to be antisymmetric if a, be R and b, ae R then a =a.

Example 1.5: Determine whether R is equivalence relation or not where

A = 0121R = 0010.11)2, 221

Solution The R is reflexive because 0, 0)1, 1).2, 2E R.

Where as R is not symmetric because 0, 1) is not in R whereas (1, 0) is in R. Hence the R is not a equivalence relation.

Closures of Relations

Sometimes when the relation R is given it may not be reflexive or transitive By adding some pairs we make the relation as reflexive or transitive. For example Letab) b, c. aab, b

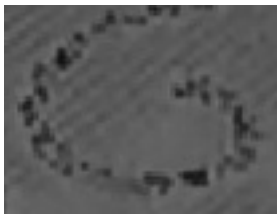


Figure 1

Revlew of Mathematical Theory

Let us discuss cach one by one -

INegation

This proposition is also called NOT proposition. It is denoted by or ~or having horizontal bar on letter.

If S is a proposition then 1S defines NOr s. The truth table for this connective is

S	1s
T	F
F	T

i) Conjunction (AND

If A and B are two propositions then conjunction of A and B is denoted by A^B. The truth table for this connective is

A	B	A^B
T	T	T
T	F	F
F	T	F
F	F	F

iii) Disjunction (OR)

If A and B are two propositions then disjunction of A and B is denoted by AvB. The truth table for this connective is

A	B	AvB
T	T.	T
T	F	T
F	T	T
F	F	F

iv) Implication

If A and B are two propositions then if A then B is a proposition. It can be denoted byABorA = B.

Revlew of Mathamatical Theory

1-8

Theory of Computation

The truth table for this connective is

AB B A

T T T

F F T

T T F

T F F

For example: Find truth value for following proposition.

If an apple is not green then it must be red"

Let, A denotes An apple is not green and B denotes It must be red"The statement A is false and B is true. Hence the proposition is true.

vf and only if If A and B are two propositions then A if and only if B. It can be denoted as AB The truth table will be

B A

AB T

T T T

F F

T

F F

F

T

For example: I will go to the school if and only if it is not raining. Here statement A is I will go to the school. The statement B is It is not raining Thus the resultant proposition is false.

1.5 Relations

One offect can be related with the other object by a some relation. Then those two objects forma pair based on this certain relationship.

Definition The relation R is a collction for the set S which represents the pair of elements.

For example a b) is in R. We can represent their relation as a R b. The first omponeo ch pai is chose rom a sel clled dmain and sod omonof each pair is chosen from a set called range.

A

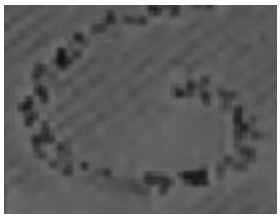


Figure 1

Review of Mathematical Theory

Theory of Computation

1-9

Propertios of Relations

A relation R on set S is

1. Reflexive if iRi for all i in S.
2. Irreflexive if iRi is false for all i in S.
3. Transitive if iRj and jRk imply iRk
4. Symmetric if iRj implies jRi.
5. Asymmetric if iRj implies that jRi is false

Every asymmetric relation must be irreflexive.

For exampleif A = a, b then

reflexive relation R can be = aabb|

irreflexive relation R can be = a, b

transitive relation R can be = a, bb, aaa

symmetric relation R can be =ab) b, a

asymmetric relation R can be =a, b)

Equivalent Relation

A relation is said to be equivalence relation if it is reflexive, symmetric and transitive, over some set S.

Suppose R is a set of relations and S is the set of elements.

For example: S is the set of lines in a plane and R is the relation of lines intersecting to each other.

A relation is said to be antisymmetric if a, b) e R and b, ae R then a =a.

Example 1.5: Determine whether R is equivalence relation or not where

A = 0121. R=/0, 010)11)2, 2)2, 1

Solution The R is reflexive because 0, 01, 1)2, 2E R Where as R is not symmetric because 0, 1) is not in R whereas 1, 0 is in R. Hence the R is not a equivalence relation.

Closures of Relations

Sometimes when the relation R is given, it may not be reflexive or transitive. By adding some pairs we make the relation as reflexive or transitive. For example Letab) b, c)(a, abb

Theory of Computation — Review of Mathematical Theory

Let us discuss each one by one -

i) Negation

This proposition is also called **NOT** proposition. It is denoted by ¬ or ~ or having horizontal bar on letter.

If S is a proposition then ¬ S defines NOT S. The truth table for this connective is

S	¬ S
T	F
F	T

ii) Conjunction (AND)

If A and B are two propositions then conjunction of A and B is denoted by A ∧ B. The truth table for this connective is

A	B	A ∧ B
T	T	T
T	F	F
F	T	F
F	F	F

iii) Disjunction (OR)

If A and B are two propositions then disjunction of A and B is denoted by A ∨ B. The truth table for this connective is

A	B	A ∨ B
T	T	T
T	F	T
F	T	T
F	F	F

iv) Implication

If A and B are two propositions then **if A then B** is a proposition. It can be denoted by A → B or A ⇒ B.

Theory of Computation — Review of Mathematical Theory

The truth table for this connective is

A	B	A → B
T	T	T
T	F	F
F	T	T
F	F	T

For example : Find truth value for following proposition.

"If an apple is not green then it must be red".

Let, A denotes "An apple is not green" and B denotes "It must be red". The statement A is false and B is true. Hence the proposition is true.

v) If and only if

If A and B are two propositions then A if and only if B. It can be denoted as A ↔ B. The truth table will be

A	B	A ↔ B
T	T	T
T	F	F
F	T	F
F	F	T

For example : "I will go to the school if and only if it is not raining". Here statement A is "I will go to the school". The statement B is "It is not raining". Thus the resultant proposition is false.

1.5 Relations

Relationship is a major aspect between two objects, even this is true in our real life. One object can be related with the other object by a some relation. Then those two objects form a pair based on this certain relationship.

Definition : The relation R is a collection for the set S which represents the pair of elements.

For example : (a, b) is in R. We can represent their relation as a R b. The first component of each pair is chosen from a set called domain and second component of each pair is chosen from a set called range.

Theory of Computation — Review of Mathematical Theory

Properties of Relations

A relation R on set S is

1. **Reflexive** if iRi for all i in S.
2. **Irreflexive** if iRi is false for all i in S.
3. **Transitive** if iRj and jRk imply iRk.
4. **Symmetric** if iRj implies jRi.
5. **Asymmetric** if iRj implies that jRi is false.

Every asymmetric relation must be irreflexive.

For example : if A = { a, b } then

reflexive relation R can be = { (a, a), (b, b) }

irreflexive relation R can be = { (a, b) }

transitive relation R can be = { (a, b), (b, a), (a, a) }

symmetric relation R can be = { (a, b), (b, a) }

asymmetric relation R can be = { (a, b) }

Equivalent Relation

A relation is said to be equivalence relation if it is **reflexive**, **symmetric** and **transitive**, over some set S.

Suppose R is a set of relations and S is the set of elements.

For example : S is the set of lines in a plane and R is the relation of lines intersecting to each other.

A relation is said to be **antisymmetric** if (a, b) ∈ R and (b, a) ∈ R then a = a.

Example 1.5

Determine whether R is equivalence relation or not where

A = { 0, 1, 2 }, R = { (0, 0), (1, 0), (1, 1), (2, 2), (2, 1) }

Solution : The R is reflexive because (0, 0), (1, 1), (2, 2) ∈ R.

Where as R is not symmetric because (0, 1) is not in R whereas (1, 0) is in R.

Hence the R is not an equivalence relation.

Closures of Relations

Sometimes when the relation R is given, it may not be reflexive or transitive.

By adding some pairs we make the relation as reflexive or transitive.

For example : Let { (a, b), (b, c), (a, a), (b, b) }